DATA ANALYTICS AND VISUALISATION | PHASE 1: PROBLEM IDENTIFICATION

Smart Air Quality and Pollution Analytics

Making India’s air-quality data easy to see, compare and act on



Problem Statement

People cannot easily monitor, understand and compare air pollution levels across Indian states and cities. A simple analytics dashboard is needed to visualise pollutant levels and identify trends and highly polluted areas.

Pollutants to visualise

AQI | **PM2.5** | **PM10** | **NO_2** | **SO_2** | **CO** | **O_3**

Monitor **Compare** **Identify hotspots**

Target Users

General Public

Check air quality in their city

Students

Learn and analyse air-pollution data

Researchers

Study pollution patterns and trends

Government Officials

Monitor levels and identify critical areas

Environmental Organizations

Analyse pollution and create awareness

City Planners

Support environmental planning and decisions

Expected Features

AQI and pollutant trends

Trends of AQI and all six pollutants

Pollution map

Colour-coded AQI by location across India

State and city comparison

Compare air quality between regions

Highly polluted areas

Top 10 ranking of the worst cities

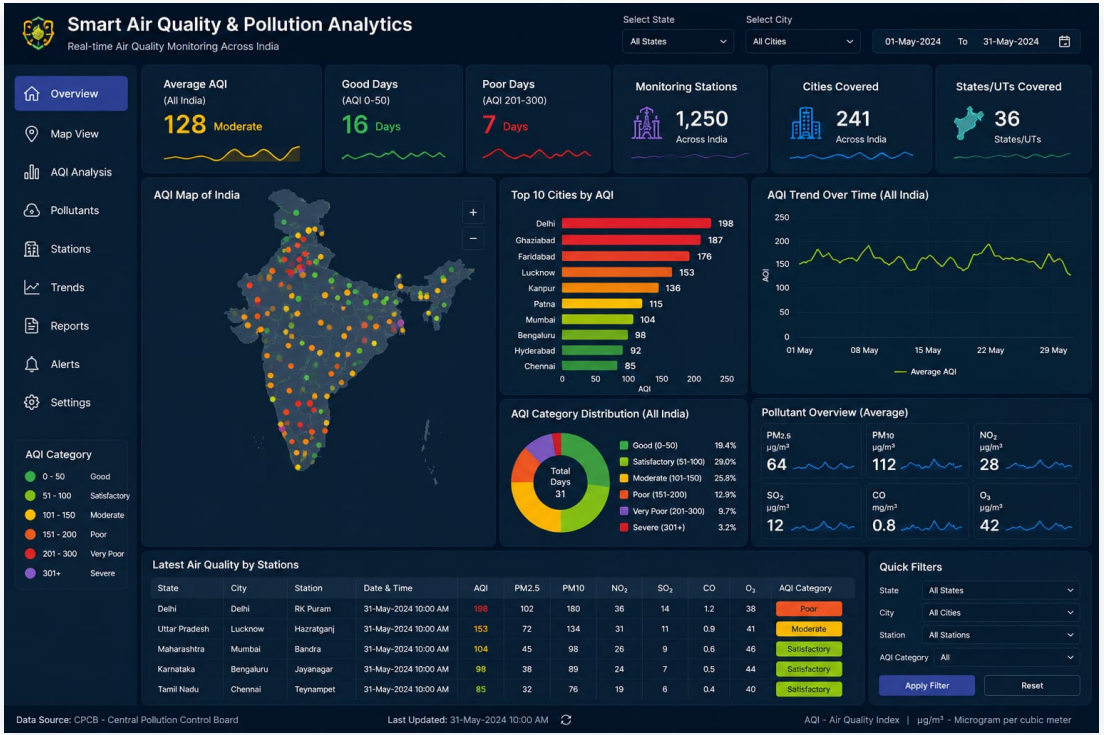
AQI prediction

Forecast near-term AQI from past trends

Monthly reports

Summaries of pollution levels by month

Expected Dashboard



At a glance

KPI cards for average AQI, good and poor days, and coverage

Explore

Map, city ranking, trend line and category split

Drill down

Filters by state, city, station and date range

Dashboard Components

KPI Cards

Headline AQI numbers

AQI Map

Spot hotspots by location

Top 10 Cities

Rank the most polluted

Trend Line

Track change over time

Category Donut

Share of days per band

Pollutant Overview

Average PM, NO_2, SO_2, CO, O_3

Station Table

Latest reading per station

Quick Filters

Slice by state, city, date

Thank You



Questions and feedback are welcome

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